

Chongyu Fan

✉ fanchon2@msu.edu 🌐 Website 🐙 Github 📄 Google Scholar (1032 citations)

Education

Michigan State University, East Lansing, USA 2024.08 – Present
Ph.D. in Computer Science
Advisor: Prof. Sijia Liu @ OPTML Lab

Huazhong University of Science and Technology, Wuhan, China 2020.09 – 2024.06
Bachelor of Engineering
Outstanding Graduate
GPA: 3.96/4.0

Industry Experience

ByteDance Seed, Student Researcher, San Jose, USA 2025.12 – Present

- **Reinforcement Pre-training (RPT) for Vision-Language Models**
 - Reframed next-token prediction into *verifiable rewards* for reinforcement learning.
 - Improved rollout efficiency by leveraging high-entropy data for exploration.
 - Achieved 8% gains with RPT over pretraining, and retained them during post-training.
- **Multi-Plan Reinforcement Learning**
 - Proposed a *multi-plan rollout framework* for reinforcement learning.
 - Improved rollout diversity, effectively mitigating *entropy collapse*.
 - Achieved 5% gains with Multi-Plan RL compared to RLVR.
- **Core Contributor for SEED 2.0**

TikTok, Research Scientist Intern, San Jose, USA 2025.05 – 2025.08

- **Reinforcement Learning for Multi-Agent Systems**
 - Identified reward imbalance issues in multi-agent reinforcement learning.
 - Proposed a *counterfactual reward* mechanism to fairly evaluate each agent’s contribution.
 - Achieved a 10% performance improvement on math reasoning tasks over state-of-the-art baselines.
- **Log-to-Leak Attack on Model Context Protocol (MCP)**
 - Discovered security issues in MCP that may lead to leakage of user-agent interaction logs.
 - Developed *Log-to-Leak*, a prompt injection attack framework for MCP.
 - Demonstrated 100% leakage success under real-world agent settings.

Research Project Highlights

• **Post-training Knowledge Editing and Alignment for LLMs and Diffusion Models**
My research focuses on developing methods to mitigate the influence of undesired knowledge in foundation models, particularly during the post-training stage of LLMs [NeurIPS’25a], [EMNLP’25] and diffusion models [ICLR’24], [NeurIPS’24a], [NeurIPS’24b], [ICLR’26c]. Through these efforts, I aim to enhance the trustworthiness, robustness, and safety of next-generation AI systems.

• **Bi-level and Smooth Optimization for LLMs and Diffusion Models**
My research leverages bi-level optimization to improve both the training and inference of LLMs and diffusion models, enhancing their robustness and interpretability [ECCV’24], and explores smooth optimization to strengthen the safety and reliability of LLMs [ICML’25], [ICLR’26b].

• **Efficient Inference and Training for Supervised, Reinforcement, and Multimodal Learning**
I investigate optimized test-time computation to enhance reasoning capability while mitigating over- and under-thinking issues [ICLR’26a], [NeurIPS’25b], design dataset condensation techniques to improve

the efficiency and scalability of reasoning model training [arXiv’25a], and study spectral optimization to accelerate vision-language-action model training and reinforcement learning [arXiv’26b].

Publications

I have published twelve papers in top-tier machine learning and computer vision venues (*e.g.*, NeurIPS, ICML, ICLR, ECCV, EMNLP), with **six** of them as first author. As of May 2026, my research has garnered **1032** citations on Google Scholar.

First-Authored Publications (* indicates equal contribution)

- [ICLR’24] C. Fan, J. Liu, Y. Zhang, D. Wei, E. Wong, S. Liu, “*SalUn: Empowering machine unlearning via gradient-based weight saliency in both image classification and generation.*” (**Spotlight, acceptance rate 5%; 400+ citations; IBM Pat Goldberg Best Paper Award Finalist**)
- [ECCV’24] C. Fan, J. Liu, A. Hero, S. Liu, “*Challenging forgets: Unveiling the worst-case forget sets in machine unlearning.*” (Travel Grant)
- [ICML’25] C. Fan, J. Jia, Y. Zhang, A. Ramakrishna, M. Hong, S. Liu, “*Towards LLM Unlearning Resilient to Relearning Attacks: A Sharpness-Aware Minimization Perspective and Beyond.*”
- [EMNLP’25] C. Fan*, C. Wang*, Y. Zhang, J. Jia, D. Wei, P. Ram, N. Baracaldo, S. Liu, “*Reasoning Model Unlearning: Forgetting Traces, Not Just Answers, While Preserving Reasoning Skills.*”
- [NeurIPS’25a] C. Fan, J. Liu, L. Lin, J. Jia, R. Zhang, S. Mei, S. Liu, “*Simplicity Prevails: Rethinking Negative Preference Optimization for LLM Unlearning.*”
- [ICLR’26a] C. Fan, Y. Zhang, J. Jia, A. Hero, S. Liu, “*CyclicReflex: Improving Reasoning Models via Cyclical Reflection Token Scheduling.*”

Co-Authored Publications

- [NeurIPS’24a] Y. Zhang, C. Fan, Y. Zhang, Y. Yao, J. Jia, J. Liu, G. Zhang, G. Liu, R. Kompella, X. Liu, S. Liu, “*UnlearnCanvas: Stylized Image Dataset for Enhanced Machine Unlearning Evaluation in Diffusion Models.*”
- [NeurIPS’24b] Y. Zhang, X. Chen, J. Jia, Y. Zhang, C. Fan, J. Liu, M. Hong, K. Ding, S. Liu, “*Defensive Unlearning with Adversarial Training for Robust Concept Erasure in Diffusion Models.*”
- [NeurIPS’25b] Y. Zhang, C. Wang, Y. Chen, C. Fan, J. Jia, S. Liu, “*The Fragile Truth of Saliency: Improving LLM Input Attribution via Attention Bias Optimization.*” (**Spotlight, acceptance rate 3%**)
- [ICLR’26b] Y. Lang, Y. Zhang, C. Fan, C. Wang, J. Jia, S. Liu, “*Downgrade to Upgrade: Optimizer Simplification Enhances Robustness in LLM Unlearning.*”
- [ICLR’26c] J. Lee, Z. Mai, J. Yoo, C. Fan, C. Zhang, W. L. Chao, “*Continual Unlearning for Text-to-Image Diffusion Models: A Regularization Perspective.*”
- [ICML’26] S. Liu, Y. Lang, S. Pal, C. Wang, Y. Huang, C. Fan, J. Diffenderfer, B. Kailkhura, Y. Zhang, “*Position: Zeroth-Order Optimization in Deep Learning Is Underexplored, Not Underpowered.*” (**Spotlight, acceptance rate 5%**)

Preprint Papers

- [arXiv’25a] J. Jia, H. Reisizadeh, **C. Fan**, N. Baracaldo, M. Hong, S. Liu, “*EPiC: Towards Lossless Speedup for Reasoning Training through Edge-Preserving CoT Condensation.*”
- [arXiv’25b] **C. Fan**, C. Wang, Y. Huang, S. Pal, S. Liu, “*LLM Unlearning Under the Microscope: A Full-Stack View on Methods and Metrics.*”
- [arXiv’26a] Y. Huang, C. Wang, **C. Fan**, Y. Lang, B. Shang, Y. Zhang, M. Hong, Q. Qu, A. Velasquez, S. Liu, “*Subspace Control: Turning Constrained Model Steering into Controllable Spectral Optimization.*”
- [arXiv’26b] **C. Fan**, G. Liu, M. Hong, R. R. Kompella, S. Liu, “*Rethinking Muon Beyond Pre-training: Spectral Failures and High-Pass Remedies for VLA and RLVR.*”

Academic Services

- **Workshop Student Co-Organizer:** New Frontiers in Adversarial ML [NeurIPS’24]
- **Reviewer:** NeurIPS’25–26, ICLR’25–26, ICML’25–26, CVPR’26, AISTATS’25

Grants & Funding

- Cisco Research Award** (\$75,000) 2025 – 2026
- Topic: “Vision–Language–Action Model Fine-Tuning through Modularity and Spectral Optimization”
 - PI: Prof. Sijia Liu
 - Role: Co-Proposal Writer

Honors

Graduate Travel Fellowship, Michigan State University	2024
Graduate Travel Fellowship, Michigan State University	2025
IBM Pat Goldberg Best Paper Award Finalist	2025
ICML’26 Gold Reviewer	2026

Mentees

- Yicheng Lang** (Undergraduate @ XJTU → Ph.D. @ MSU) – [ICLR’26b] 2025.05 – 2025.09